

Notes: Cui, Wright, & Zhu (JPE, 2025)

Endogenous Liquidity and Capital Reallocation

Contents

1	Big picture	3
2	Data and measurement (macro facts, micro facts, and calibration targets)	3
2.1	Macro data and the core reallocation objects	3
2.2	Macro empirical patterns the model is built to explain	4
2.3	Micro evidence on money holdings and reallocation	4
2.4	Micro evidence on money holdings and inflation	4
2.5	Calibration targets used later in the quantitative section	5
3	Models and empirical specifications (core equations + interpretation)	5
3.1	Environment and production side	5
3.2	The centralized-market problem	6
3.3	The decentralized market: trade, surpluses, and liquidity constraints	6
3.4	Bargaining and the full-versus-partial sale margin	6
3.5	Market clearing and effective capital	7
3.6	Steady-state IS-LM system and normalized liquidity	7
3.7	Efficiency and the optimal capital subsidy	8
3.8	Business-cycle shock process	8
4	Identification, calibration, and what makes the quantitative exercise credible	8
4.1	This is not a quasi-experimental paper	8
4.2	What is structurally identified or disciplined?	9
4.3	Why the quantitative exercise is convincing	9
4.4	Main limitations	9
5	Tables and figures	10
5.1	Figure 1: Inflation and real returns on different assets	10
5.2	Figure 2: Inflation and reallocation	10
5.3	Figure 3 and table 1: Debt, investment, and reallocation over the business cycle . . .	11

5.4	Tables 2–4: Micro evidence on firm cash and reallocation	11
5.5	Figure 4: Full and partial sales in DM trade	12
5.6	Figure 5 and table 5: Steady state and comparative statics	12
5.7	Table 6: Calibration	13
5.8	Figures 6–8: Long-run and medium-run effects of inflation	13
5.9	Tables 7–8: Short-run dynamics and the role of money	14
5.10	Figures 9–11: Search frictions, fiscal policy, and persistent shocks	14
6	Short summary	15
7	Conclusion	16
7.1	Core conclusion (what the paper establishes)	16
7.2	Economic intuition	16
7.3	Takeaway	16

1 Big picture

- **Question.** How do liquidity frictions shape capital reallocation across firms, and how does that interact with inflation, credit conditions, and aggregate business-cycle dynamics?
- **Core novelty.** The paper distinguishes **full sales** (the buyer acquires all of the seller’s capital) from **partial sales** (the buyer acquires only part of it). Under constant returns to scale, full sales are socially efficient, so partial sales are a direct symptom of liquidity frictions.
- **Empirical facts.** In the long run, higher inflation is associated with less total reallocation, fewer full sales, and more partial sales. At business-cycle frequencies, the signs reverse: inflation comoves positively with reallocation and negatively with the partial-sale share.
- **Theory.** Firms buy capital in a centralized primary market and then retrade it in a decentralized secondary market after idiosyncratic productivity shocks. Secondary-market trade features search, bilateral bargaining, and liquidity constraints coming from both money and limited credit.
- **Key mechanism.** High inflation raises the cost of holding liquid assets, so buyers enter secondary trade with less purchasing power. That depresses reallocation and makes partial sales more common. In the short run, easier credit both raises reallocation and lowers money demand, so inflation can rise at the same time that reallocation rises.
- **Quantitative headline.** A model with aggregate productivity shocks alone gets the reallocation facts badly wrong; adding credit shocks allows the model to match the long-run, medium-run, and business-cycle behavior of reallocation while still fitting standard RBC moments.
- **Why the paper matters.** The paper ties together monetary economics, search, and capital reallocation. It shows that “liquidity” is not just about household transactions: it also governs whether productive firms can buy capital from less productive firms, with implications for TFP, welfare, and policy.

2 Data and measurement (macro facts, micro facts, and calibration targets)

2.1 Macro data and the core reallocation objects

- **Sample.** U.S. data, 1971–2018 for the empirical facts; 1984–2018 for calibration because the key reallocation ratios look much more stable after 1984.
- **Main source.** Compustat. The paper constructs aggregate series from firm-level information on acquisitions and sales of property, plant, and equipment, together with total capital expenditures.
- **Reallocation.** Defined as full sales plus partial sales.
- **R share.** Reallocation divided by total capital expenditure, where total capital expenditure means new investment plus reallocation. This measures how important used-capital trade is relative to all capital spending.
- **P share.** Partial sales divided by total reallocation. This measures the composition of reallocation.
- **Levels.** After 1984, the R share fluctuates around roughly 32%, while the P share fluctuates around roughly 24%.

- **Cost of liquidity.** Measured mainly with CPI inflation, although nominal T-bill and AAA corporate bond rates give similar patterns.
- **Credit conditions.** Proxied with aggregate firm debt in the data and with pledgeability/credit parameters in the model.

2.2 Macro empirical patterns the model is built to explain

- **Long run.** Higher inflation is associated with lower R share and higher P share. Intuitively, costly liquidity suppresses full acquisitions more than partial trades.
- **Business cycle.** The cyclical component of the R share is **procyclical**; the cyclical component of the P share is **countercyclical**. Debt is procyclical and comoves positively with reallocation.
- **Relative volatility.** Reallocation is more volatile than investment, which is exactly what is needed to generate a procyclical R share.
- **Additional stylized facts.** The paper also wants to match procyclical used-capital prices, countercyclical productivity dispersion, and the procyclicality of spending on used capital relative to total investment.

2.3 Micro evidence on money holdings and reallocation

- **Firm liquidity measure.** Cash plus cash equivalents, including assets readily convertible into cash.
- **Full purchases: extensive margin.** The paper estimates linear probability models where the dependent variable is an indicator for engaging in a full purchase this year.
- **Full purchases: intensive margin.** It also regresses log full-purchase spending on lagged money holdings among firms that do engage in full purchases.
- **Controls.** Lagged EBIT, total assets, leverage, and relative size, with firm fixed effects and either year or year-by-industry fixed effects.
- **Main micro result 1.** In table 2, a 1% increase in money holdings raises the probability of a full purchase by about 0.00018 in levels, which is about 0.1% relative to the average full-sale probability.
- **Main micro result 2.** The spending elasticity is about 0.19–0.20: a 1% increase in money holdings raises full-purchase spending by roughly 0.2%.
- **State-level aggregation.** Because partial-purchase buyers are not directly observed, the paper aggregates to the state level and shows that higher state-level money holdings raise total reallocation and lower the partial-sale share.
- **State-level magnitudes.** In table 3, the elasticity of reallocation with respect to money holdings is about 0.21–0.28, while the elasticity of the P share is about -0.36 to -0.47 .

2.4 Micro evidence on money holdings and inflation

- **National inflation.** In table 4, a 1% increase in inflation lowers firm money holdings by about 1.27%.

- **Cyclical inflation.** Using the cyclical component of inflation, the elasticity is even larger in magnitude, around -2.71 .
- **Cross-state inflation.** Using state-level inflation to exploit cross-sectional variation, the paper again finds significantly negative effects, with coefficients around -2.09 and -1.75 .
- **Interpretation.** These regressions are important because they directly link inflation to firms' liquid balances, and liquid balances to full versus partial reallocation.

2.5 Calibration targets used later in the quantitative section

- **Standard macro targets.** Hours, investment/output, depreciation, tax rates, government share, and real and nominal yields.
- **Capital-reallocation targets.** The average R share (0.32), average P share (0.24), firm money holdings over output (4.2%), and the elasticity of acquisition spending with respect to inflation (-0.64).
- **Shock discipline.** In business-cycle exercises, productivity-shock volatility is chosen to match output volatility; adding credit shocks lets the model also match the volatility of the R share.

3 Models and empirical specifications (core equations + interpretation)

3.1 Environment and production side

Each period has a frictionless centralized market (CM) followed by a frictional decentralized market (DM). Firms buy capital in the CM and then retrade it in the DM after idiosyncratic productivity shocks. To avoid notational clash, I write the labor-share parameter as η .

$$U(c, 1 - h) = u(c) + \psi(1 - h), \quad 0 < \beta < 1, \quad (1)$$

and the production function is

$$f(k, h) = (A\varepsilon k)^{1-\eta} h^\eta. \quad (2)$$

After static labor demand is solved out, profit becomes linear in effective capital:

$$\Pi(k, \varepsilon) = B(w) \varepsilon k, \quad B(w) = \left(\frac{\eta}{w} \right)^{\eta/(1-\eta)} (1 - \eta) A. \quad (3)$$

- **Interpretation.** Linearity in εk is the key reason full reallocation is efficient under constant returns to scale. If a buyer has higher productivity than a seller, efficiency says the buyer should get *all* of the seller's capital.
- **Economic intuition.** Partial sales are not efficient smoothing in the benchmark model; they arise because the buyer does not have enough liquidity to buy the whole bundle.

3.2 The centralized-market problem

Let W be the CM value function, Ω beginning-of-market wealth, t_h the labor tax, and t_k the capital-income tax. Then

$$W(\Omega, \varepsilon) = \max_{c, h, \hat{k}, \hat{m}} \left\{ u(c) + \psi(1 - h) + \beta \mathbb{E}_{\varepsilon|\varepsilon} V_1(\hat{k}, \hat{z}, \hat{\varepsilon}) \right\} \quad (4)$$

subject to

$$c = \Omega + (1 - t_h)wh - \phi\hat{m} - \hat{k}, \quad \hat{z} = \phi_1\hat{m}. \quad (5)$$

Because $\Omega = (1 - t_k)B(w)\varepsilon k + (1 - \delta)k + z - d - T$, the first-order conditions are

$$\frac{\psi}{(1 - t_h)w} = \beta \mathbb{E}_{\varepsilon|\varepsilon} \left[\frac{\partial V_1(\hat{k}, \hat{z}, \hat{\varepsilon})}{\partial \hat{k}} \right], \quad \frac{\psi}{(1 - t_h)w} \frac{\phi}{\phi_1} = \beta \mathbb{E}_{\varepsilon|\varepsilon} \left[\frac{\partial V_1(\hat{k}, \hat{z}, \hat{\varepsilon})}{\partial \hat{z}} \right]. \quad (6)$$

- **Interpretation.** Firms choose next period's capital and liquid balances by equating the current resource cost to the expected marginal value in next period's DM.
- **Important simplification.** The CM value function is linear in wealth, so the model stays tractable even with heterogeneous idiosyncratic shocks.

3.3 The decentralized market: trade, surpluses, and liquidity constraints

When two firms meet in the DM, the high-productivity firm is the buyer and the low-productivity firm is the seller. Let q be capital transferred, p the cash payment, and d the promised CM payment. Buyer and seller surpluses are

$$S^b(s, \tilde{s}) = \frac{\psi([(1 - t_k)\varepsilon B(w) + (1 - \delta)]q - p - d)}{w(1 - t_h)}, \quad (7)$$

$$S^s(\tilde{s}, s) = \frac{\psi(p + d - [(1 - t_k)\tilde{\varepsilon} B(w) + (1 - \delta)]q)}{w(1 - t_h)}. \quad (8)$$

Liquidity comes from cash and limited credit. The debt limit is

$$D = x_0 + x_\Pi \Pi + x_q(1 - \delta)q + x_k(1 - \delta)k. \quad (9)$$

- **Interpretation of the credit terms.** x_0 is unsecured debt, x_Π is debt backed by future profit, x_q is debt backed by the newly purchased capital, and x_k is debt backed by existing capital.
- **Why this matters.** The composition of liquidity determines whether the buyer can purchase the entire asset bundle or must settle for a partial sale.

3.4 Bargaining and the full-versus-partial sale margin

The DM terms of trade are pinned down by Kalai proportional bargaining:

$$\max_{d,p,q} S^b(s, \tilde{s}) \quad \text{s.t.} \quad (1 - \nu)S^b(s, \tilde{s}) = \nu S^s(\tilde{s}, s), \quad q \leq \tilde{k}, \quad p \leq z, \quad d \leq D, \quad (10)$$

where ν is buyer bargaining power.

The paper derives a productivity threshold

$$\bar{\varepsilon} = \Psi_0 - \Psi_1 \tilde{\varepsilon}, \quad \Psi_0 = \frac{z + x_0 + x_k(1 - \delta)k/\tilde{k} - (1 - \delta)(1 - x_q)}{(1 - t_k)[1 - \nu - x_\Pi(1 + k/\tilde{k})]B(w)}, \quad \Psi_1 = \frac{\nu}{1 - \nu - x_\Pi(1 + k/\tilde{k})}. \quad (11)$$

When the buyer is constrained, trade is partial and

$$Q = \frac{z + x_0 + [(1 - \delta)x_k + (1 - t_k)B(w)x_\Pi\varepsilon]k}{(1 - t_k)B(w)[(1 - \nu - x_\Pi)\varepsilon + \nu\tilde{\varepsilon}] + (1 - \delta)(1 - x_q)} < \tilde{k}. \quad (12)$$

- **Economic meaning of proposition 1.** Whether a meeting ends in a full or partial sale depends on the buyer's productivity *and* available liquidity.
- **Crucial comparative static.** Higher money balances z shift the threshold in favor of full sales. So more liquidity does not just increase how much capital changes hands; it changes *the form* of reallocation.
- **Deep intuition.** If profit-backed credit (x_Π) is weak, the most productive buyers can actually be the most constrained because they value the asset the most but cannot fully finance it.

3.5 Market clearing and effective capital

Money and goods market clearing are

$$\phi_1 M_1 = \int \hat{z}(\varepsilon) dF(\varepsilon), \quad c + K_1 + G = Y + (1 - \delta)K, \quad (13)$$

with effective post-reallocation capital given by

$$K = a \iint_{\varepsilon > \tilde{\varepsilon}} \varepsilon [k + q(s, \tilde{s})] d\Gamma(\tilde{s}) d\Gamma(s) + a \iint_{\varepsilon < \tilde{\varepsilon}} \varepsilon [k - q(\tilde{s}, s)] d\Gamma(\tilde{s}) d\Gamma(s) + (1 - a) \int \varepsilon k d\Gamma(s). \quad (14)$$

- **Interpretation.** Aggregate capital is not enough; what matters is *who* holds it after trade. Inflation can lower output by reducing this reallocation-adjusted effective capital stock.

3.6 Steady-state IS-LM system and normalized liquidity

Define normalized liquidity

$$L = \frac{(Z + x_0)/K - (1 - \delta)(1 - x_q - x_k)}{(1 - t_k)B(w)}. \quad (15)$$

Then the steady state is characterized by an IS curve, an LM curve, and the resource constraint:

$$\frac{r + \delta}{(1 - t_k)B(w)} = \mathbb{E}[\varepsilon] + a(1 - \nu)I_s(L) + (1 - \delta)x_k I_{b1}(L) + x_\Pi(1 - t_k)B(w)a\nu I_{b2}(L), \quad (16)$$

$$i = a\nu \iint_{S^b(L)} \frac{\varepsilon - \tilde{\varepsilon}}{(1 - \nu - x_\Pi)\varepsilon + \nu\tilde{\varepsilon} + (1 - \delta)(1 - x_q)/[(1 - t_k)B(w)]} dF(\tilde{\varepsilon})dF(\varepsilon), \quad (17)$$

$$(u')^{-1}\left(\frac{\psi}{(1 - t_h)w}\right) + G = \left(\frac{B(w)J(L, w)}{1 - \eta} - \delta\right) K. \quad (18)$$

- **Interpretation.** These equations are the model's structural backbone. The first is a capital Euler equation, the second is money demand, and the third is goods-market clearing.
- **Why they are useful.** The paper can study equilibrium existence, uniqueness, and comparative statics using the geometry of these curves in (L, B) space.

3.7 Efficiency and the optimal capital subsidy

Even if the Friedman rule is approached, bargaining creates a hold-up problem for primary investment. The efficient corrective capital tax/subsidy is

$$t_k^* = 1 - \frac{\int_0^\infty \hat{\varepsilon} dF(\hat{\varepsilon}) + a \iint_{\hat{\varepsilon} < \tilde{\varepsilon}} (\tilde{\varepsilon} - \hat{\varepsilon}) dF(\tilde{\varepsilon})dF(\hat{\varepsilon})}{\int_0^\infty \hat{\varepsilon} dF(\hat{\varepsilon}) + a(1 - \nu) \iint_{\hat{\varepsilon} < \tilde{\varepsilon}} (\tilde{\varepsilon} - \hat{\varepsilon}) dF(\tilde{\varepsilon})dF(\hat{\varepsilon})} \leq 0. \quad (19)$$

- **Interpretation.** The model wants a capital *subsidy*, not a tax, because sellers do not internalize the full social value of bringing capital into the DM.
- **Important message.** Even in a monetary model, the Friedman rule need not be globally optimal once bargaining and fiscal distortions are present.

3.8 Business-cycle shock process

For short-run quantitative work, the aggregate state follows

$$\ln A_t = \rho_A \ln A_{t-1} + \varsigma_{A,t}, \quad \ln x_{q,t} - \ln x_q = \rho_x (\ln x_{q,t-1} - \ln x_q) + \varsigma_{x,t}, \quad (20)$$

with orthogonal innovations $\varsigma_{A,t}$ and $\varsigma_{x,t}$.

- **Interpretation.** Productivity shocks alone generate standard RBC fluctuations, but the paper's main claim is that reallocation facts require liquidity or credit shocks as well.

4 Identification, calibration, and what makes the quantitative exercise credible

4.1 This is not a quasi-experimental paper

- **No IV or natural experiment.** The macro part of the paper is not trying to estimate a causal reduced-form effect of inflation on reallocation using external variation.

- **What the evidence does instead.** The empirical sections document robust patterns: inflation covaries with reallocation composition in different ways across frequencies; firm cash holdings predict full versus partial purchases; and inflation predicts lower firm cash holdings.
- **Role of theory.** The model is used to explain why those reduced-form correlations look different in the long run and at business-cycle frequencies.

4.2 What is structurally identified or disciplined?

- **Externally pinned parameters.** The nominal AAA yield gives $i = 0.067$, average inflation implies $\beta \approx 0.962$, hours determine ψ , and the investment-output ratio plus depreciation discipline η and δ .
- **Productivity dispersion.** The idiosyncratic productivity parameter $\sigma_\varepsilon = 1.2853$ is backed out from Compustat productivity dispersion after adjusting for the capital share.
- **DM-entry distribution.** The mean and variance of the entry-cost distribution, (μ_g, σ_g) , are chosen jointly with the liquidity parameters.
- **Liquidity parameters.** x_q and x_Π are calibrated to match four long-run moments: the average R share, average P share, firm money over output, and the elasticity of acquisition spending with respect to inflation.
- **Shock volatilities.** With only productivity shocks, σ_A is chosen to match output volatility. With both productivity and credit shocks, (σ_A, σ_x) are chosen to match output volatility and R share volatility.

4.3 Why the quantitative exercise is convincing

- **Micro-to-macro consistency.** The same mechanism is supported in the cross section and in the aggregate time series: money holdings facilitate full purchases, while inflation reduces money holdings.
- **Non-targeted moments.** After calibration, the model matches not only standard RBC facts but also the signs and magnitudes of the correlations between output and the R share, P share, inflation, used-capital prices, and productivity dispersion.
- **Counterfactual comparison.** The paper shows explicitly that a model with only productivity shocks fails, and that a nonmonetary credit model also does materially worse on the composition of reallocation.
- **Analytical structure.** The model is not a black box. Many comparative statics and policy results are available analytically, which makes the calibration easier to interpret.

4.4 Main limitations

- **Inflation as the cost of liquidity.** This is plausible and empirically motivated, but it is still a proxy. The paper itself emphasizes that many assets provide liquidity, not just cash.
- **Reduced-form credit shocks.** The short-run credit process enters through a collateral or pledgeability parameter rather than an explicit banking balance-sheet model.

- **Measurement of partial sales.** Aggregate full and partial capital reallocation are built from Compustat items that inevitably miss some private transactions, rentals, and smaller firms.
- **Frequency decomposition.** The interpretation of long-run versus cyclical patterns relies on band-pass filtering, so some details can vary with the filter window even if the qualitative message is robust.

5 Tables and figures

5.1 Figure 1: Inflation and real returns on different assets

Read:

- Real returns on transaction deposits, T-bills, long Treasury bonds, mortgages, and corporate bonds are all strongly negatively correlated with inflation.
- This is a key motivation for using inflation as the cost of liquidity. The point is not just that inflation taxes cash directly, but that it depresses the return on a wide set of liquid or near-liquid assets.
- So the paper’s monetary mechanism should be read broadly: inflation worsens the terms at which firms convert current resources into future purchasing power.

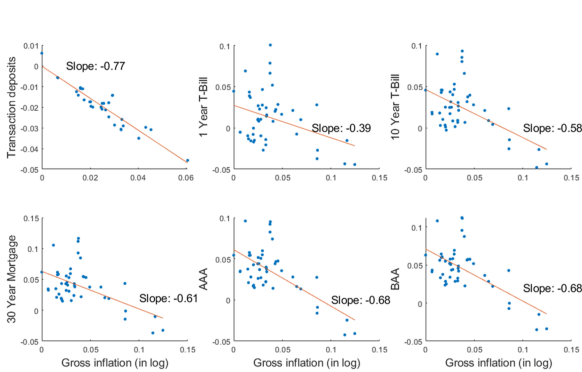


FIG. 1.—Inflation and real returns on different assets.

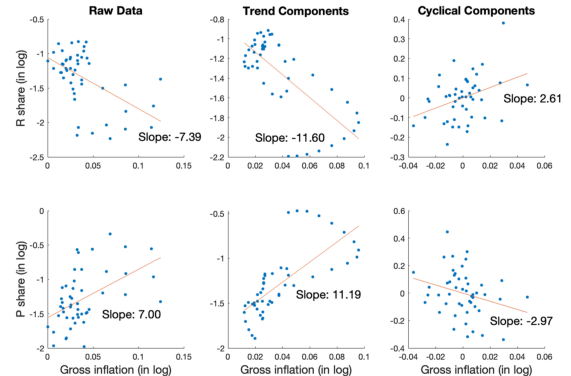


FIG. 2.—Inflation and reallocation. $R^2 = 0.28, 0.52, 0.13$ (first row); $0.25, 0.54, 0.07$ (second row). p -values for slopes = $1.1 \times 10^{-4}, 7.2 \times 10^{-3}, 0.01$ (first row); $2.9 \times 10^{-4}, 3.3 \times 10^{-5}, 0.06$ (second row).

5.2 Figure 2: Inflation and reallocation

Read:

- In raw and trend components, higher inflation is associated with lower R share and higher P share.
- In cyclical components, the signs reverse: inflation is positively related to the R share and negatively related to the P share.
- This is the central empirical pattern the paper wants to explain. Long-run inflation looks like a liquidity tax; short-run inflation partly reflects easier credit and lower money demand.

5.3 Figure 3 and table 1: Debt, investment, and reallocation over the business cycle

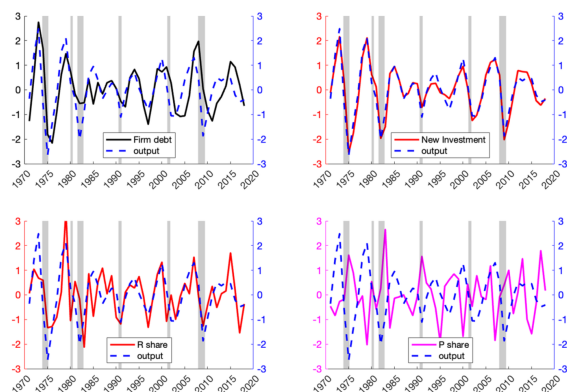


FIG. 3.—Debt, investment, and reallocation over business cycle. Each series is normalized by its own standard deviation. Shaded areas denote National Bureau of Economic Research recession dates.

TABLE 1
BUSINESS CYCLE STATISTICS

Correlation	Debt	R Share	P Share	Investment	Inflation	Output
Debt	1	.53	-.44	.62	.42	.64
R share	...	1	-.77	.62	.36	.62
P share	...	...	1	-.50	-.27	-.52
Investment	...	...	...	1	.43	.96
Inflation	...	...	...	...	1	.33
Output	...	...	...	...	...	1
Relative standard deviation	1.41	6.44	9.66	2.58	.89	1

NOTE.—Output is private output defined as the sum of consumption and investment. Its standard deviation is 1.73%.

Read:

- Debt and the R share are procyclical, while the P share is countercyclical.
- Table 1 shows that the correlation of debt with the R share is 0.53, while the P share is negatively correlated with debt and output.
- Reallocation is much more volatile than output or investment, which matters because the model needs that extra volatility to generate cyclical movement in reallocation shares.

5.4 Tables 2–4: Micro evidence on firm cash and reallocation

TABLE 2
FULL SALES AND MONEY HOLDINGS

	PROBABILITY			SPENDING		
	(1)	(2)	(3)	(4)	(5)	(6)
Money holdings	.018*** (.001)	.019*** (.001)	.018*** (.001)	.190*** (.012)	.199*** (.012)	.202*** (.013)
EBIT	.024*** (.002)	.027*** (.002)	.028*** (.002)	.270*** (.019)	.274*** (.019)	.266*** (.020)
Asset	.087*** (.003)	.066*** (.003)	.069*** (.003)	-.322*** (.018)	-.390*** (.022)	-.381*** (.025)
Leverage	-.001** (.000)	-.001** (.000)	-.001* (.000)	-.005 (.003)	-.005 (.003)	-.005* (.003)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects		Yes			Yes	
Year × industry fixed effects			Yes			Yes
Adjusted R^2	.029	.039	.053	.049	.075	.112
Observations	115,822	115,822	115,822	33,678	33,678	33,678

NOTE.—The table reports results from regressing whether a firm engages in a full purchase and full purchase spending on its money holdings. Robust standard errors (in parentheses) are clustered at the firm level. All independent variables are lagged by one period. Acquisition spending, money holdings, and EBIT are normalized by firms' total assets.

* $p < .10$.
** $p < .05$.
*** $p < .01$.

TABLE 3
MONEY HOLDINGS ON REALLOCATION: STATE-LEVEL EVIDENCE

	REALLOCATION		P SHARE	
	(1)	(2)	(3)	(4)
Money holdings	.282*** (.071)	.205** (.087)	-.470*** (.079)	-.350*** (.097)
EBIT	-.061 (.101)	-.001 (.097)	.034 (.071)	-.113 (.070)
Asset	-.098 (.061)	-.186** (.080)	-.395*** (.070)	-.042 (.076)
Leverage	-.145 (.130)	-.149 (.146)	-.022 (.107)	-.135 (.140)
Year fixed effects		Yes		Yes
Adjusted R^2	.054	.141	.196	.304
Observations	2,312	2,312	2,297	2,297

NOTE.—The table reports results from regressing reallocation and P share on money holdings at the state level. Robust standard errors (in parentheses) are clustered at the state level. Reallocation, money held, and EBIT are normalized by total assets.

** $p < .05$.
*** $p < .01$.

Read:

- Table 2 shows that firms with more money holdings are more likely to make full purchases and spend more when they do so.
- Table 3 shows the same mechanism at the state level: more money is associated with more total reallocation and fewer partial sales.

- Table 4 links the mechanism back to inflation: higher inflation reduces firms' liquid balances. This closes the reduced-form chain *inflation* $\rightarrow$ *less money* $\rightarrow$ *fewer full purchases / more partial purchases*.

5.5 Figure 4: Full and partial sales in DM trade

TABLE 4
MONEY HOLDINGS AND LIQUIDITY COST

	MONEY HOLDINGS			
	(1)	(2)	(3)	(4)
Inflation	-1.268*** (.170)	...	...	...
Inflation (cycle)	...	-2.706*** (.160)	...	...
State-level inflation	...	...	-2.092*** (.524)	-1.753*** (.534)
EBIT	.122*** (.005)	.119*** (.005)	.108*** (.007)	.107*** (.007)
Asset	-.091*** (.005)	-.085*** (.006)	-.085*** (.013)	-.082*** (.013)
Leverage	-.000*** (.000)	-.000*** (.000)	-.001* (.001)	-.001** (.000)
Capital expenditure	-.102*** (.005)	-.105*** (.006)	-.088*** (.008)	-.078*** (.008)
Firm fixed effects	Yes	Yes	Yes	Yes
Year fixed effects				
Year $\times$ industry fixed effects				Yes
Adjusted R^2	.013	.015	.029	.056
Observations	142,303	142,303	91,851	91,851

NOTE.—The table reports results from regressing money holdings on inflation rates. Robust standard errors (in parentheses) are clustered at the firm level. Money holdings, EBIT, and capital expenditures are normalized by total assets.

* $p < .10$.
 ** $p < .05$.
 *** $p < .01$.

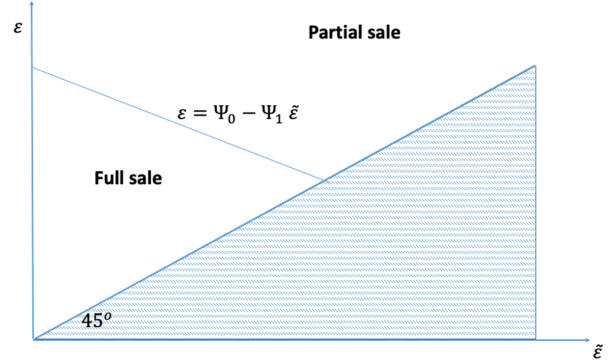


FIG. 4.—Full and partial sales in DM trade.

Read:

- The figure plots the threshold $\varepsilon = \Psi_0 - \Psi_1 \tilde{\varepsilon}$ that divides full from partial sales.
- Below the threshold, the buyer can finance the entire acquisition and there is a full sale. Above it, the buyer is liquidity constrained and only a partial sale occurs.
- The geometry is useful because it makes the paper's main theoretical point visual: more liquidity expands the full-sale region.

5.6 Figure 5 and table 5: Steady state and comparative statics

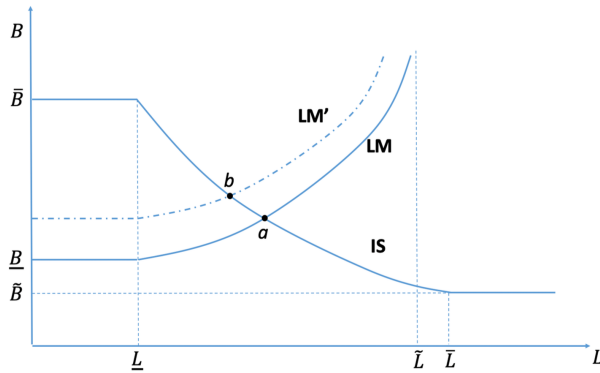


FIG. 5.—Steady state.

TABLE 5
COMPARATIVE STATICS

	ι	χ_0	χ_k	χ_q	χ_{π}	τ_k	τ_h	G	A	ι^*	ι^b
K	$\pm$	0	+	$\pm$	$\pm$	-	-	+	+	$\pm$	+
Z	$\pm$	-	$\pm$	$\pm$	$\pm$	-	-	+	+	$\pm$	$\pm$
w	$\pm$	0	+	+	+	-	0	0	+	-	+
c	$\pm$	0	+	+	+	-	-	0	+	-	+
H	$\pm$	0	$\pm$	$\pm$	$\pm$	-	-	+	$\pm$	$\pm$	$\pm$
Y	$\pm$	0	+	$\pm$	$\pm$	-	-	+	+	$\pm$	+
Φ	-	0	+	+	$\pm$	-	0	0	0	-	-

NOTE.—All effects assume that χ_q is big and χ_s is small.

^a Assumes that χ_k is small.

^b Assumes that χ_k and θ are big.

Read:

- Figure 5 rewrites the model in IS-LM language. The steady state is the intersection of an investment condition and a liquidity condition in (L, B) space.
- Table 5 shows that many effects are nonmonotone. Inflation can either raise or lower capital because money and capital are both substitutes and complements in liquidity services.
- Some signs are robust and especially important: higher inflation lowers the probability of full sales, while relaxing collateral limits raises it.

5.7 Table 6: Calibration

Read:

- The calibrated benchmark uses $i = 0.067$, $\beta = 0.9619$, $\delta = 0.10$, bargaining power $\nu = 0.50$, and liquidity parameters $x_q = 0.8869$, $x_\Pi = 0.0989$.
- The long-run capital-allocation targets are exactly the moments one would hope to see in this paper: R share, P share, firm money/output, and acquisition elasticity with respect to inflation.
- This table is helpful because it shows the model is not being calibrated only to generic RBC moments.

TABLE 6 CALIBRATED PARAMETER VALUES		
Parameter	Value	Targets
i	.067	Nominal AAA yield
β	.9619	Real AAA yield
ξ	2.2478	Labor hours
η	.611	Investment/output
δ	.1	Depreciation rate
σ_e	1.2853	Compustat
σ_γ	.2217	Acquisition elasticity
μ_γ	-1.7684	R share
χ_q	.8869	P share
χ_Π	.0989	Cash/output
τ_k	.25	Capital tax rate
τ_h	.22	Labor tax rate
G	.1471	Government share
θ	.50	Symmetry

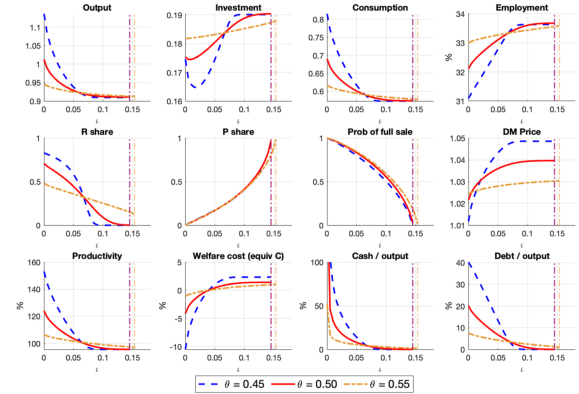


FIG. 6.—Long-run effects of inflation. Vertical lines divide nonmonetary (left) and monetary (right) regions. For productivity, the level in the benchmark calibration is used as the normalization. For welfare, the level at zero inflation is used as the normalization.

5.8 Figures 6–8: Long-run and medium-run effects of inflation

Read:

- Figure 6 shows that higher long-run inflation reduces output, consumption, cash/output, debt/output, the R share, and the probability of a full sale, while raising the P share.
- Welfare losses are economically meaningful. At the benchmark bargaining power, moving from 10% inflation to price stability is worth roughly 1.4% of consumption, and moving all the way to the Friedman rule is worth roughly 5.5%.
- Figures 7 and 8 compare the model with long-run and medium-run data. The paper is not claiming a literal one-factor explanation of the entire post-1970 period, but the inflation-based mechanism tracks the broad rise in full sales and decline in partial sales remarkably well.

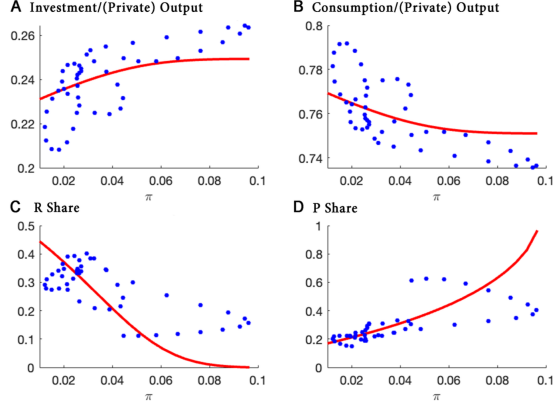


FIG. 7.—Long-run effects in data and model. Circles are scatterplots of data. To capture the long run, we use the previous trend components (frequencies above 9 years) from the band-pass filter. The results are robust with different specification. Solid lines are model predictions.

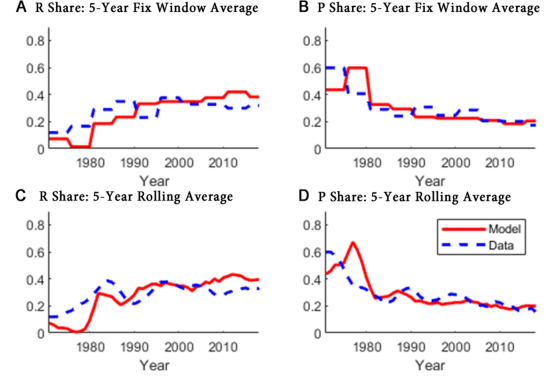


FIG. 8.—Medium-run effects in data and model. The figure shows the R share and P share from the model (solid lines) and data (dashed lines). A and B are obtained by taking the average in fixed 5-year windows. Therefore, the time series are constant within each 5-year window. C and D are calculated by taking 5-year rolling averages.

5.9 Tables 7–8: Short-run dynamics and the role of money

TABLE 7
BUSINESS CYCLE STATISTICS

	STANDARD DEVIATION			CORRELATION WITH OUTPUT		
	Data (1)	A Only (2)	A and χ_q (3)	Data (4)	A Only (5)	A and χ_q (6)
Output	1.73	1.73	1.73	1.00	1.00	1.00
Consumption	.66	.44	.58	.96	.97	.89
Investment	2.58	2.89	2.73	.96	.99	.95
Employment	.88	.45	.76	.85	.97	.92
Total factor productivity	.69	.62	.48	.76	1.00	.92
R share	6.44	1.78	6.44	.62	-.99	.47
P share	9.66	1.17	7.45	-.52	.98	-.55
Inflation	.89	.39	1.18	.33	-.76	.42

NOTE.—Standard deviations of other variables are relative to output.

TABLE 8
BUSINESS CYCLE STATISTICS WITHOUT MONEY

	STANDARD DEVIATION			CORRELATION WITH OUTPUT		
	Data (1)	A Only (2)	A and χ_q (3)	Data (4)	A Only (5)	A and χ_q (6)
Output	1.73	1.73	1.73	1.00	1.00	1.00
Consumption	.66	.43	.51	.96	.96	.92
Investment	2.58	2.93	2.81	.96	.99	.97
Employment	.88	.43	.66	.85	.97	.90
Total factor productivity	.69	.63	.54	.76	1.00	.94
R share	6.44	2.45	6.44	.62	-.97	.25
P share	9.66	1.16	4.91	-.52	.96	-.37
Inflation	.89	...	...	.33	...	...

NOTE.—Standard deviations are relative to output, except for output itself.

Read:

- With only aggregate productivity shocks, the model fails badly on reallocation dynamics: the R share is countercyclical in the model ($\rho = -0.99$) even though it is procyclical in the data ($\rho = 0.62$), and the P share moves with the wrong sign.
- Adding credit shocks fixes this. In table 7, the model with A and χ_q shocks gets the R share correlation (0.47) and P share correlation (-0.55) close to the data (0.62, -0.52).
- Table 8 shows why money matters. A pure-credit model improves on the one-shock benchmark but still understates P share volatility and produces too weak a positive correlation between output and the R share.

5.10 Figures 9–11: Search frictions, fiscal policy, and persistent shocks

Read:

- Figure 9 shows that better search ($a \uparrow$) raises output and welfare because firms can rely more on secondary trade and consume more in the primary market. Search frictions are quantitatively important.
- Figure 10 shows that a capital tax reduces reallocation and money demand. The model likes a substantial capital subsidy in welfare terms because bargaining leads to underinvestment.

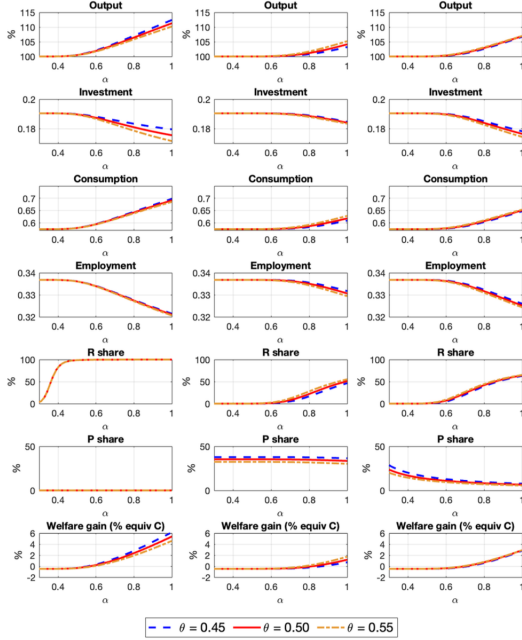


FIG. 9.—Long-run effects of search frictions. The first column is for the case of perfect credit, the second is for imperfect credit and no money, and the third is for imperfect credit and money. Output when $\alpha = 0$ is normalized to 1. Welfare gain is calculated relative to the calibrated economy in the main text.

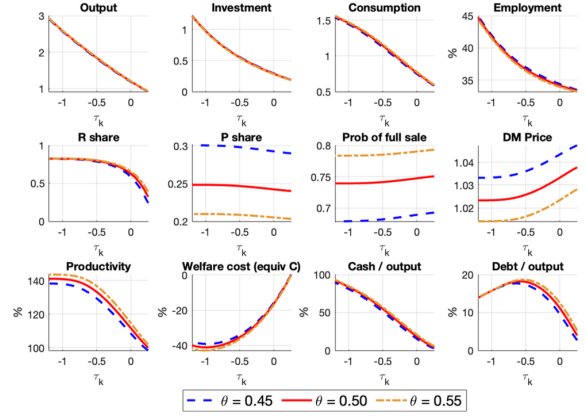


FIG. 10.—Long-run effects of capital taxation.

- Figure 11 adds persistent firm heterogeneity. Larger persistent productivity gaps make big firms hold more capital, often less cash, and become more likely to swallow small firms in full sales. The P share rises because heterogeneity makes financing the efficient transfer harder.

6 Short summary

- **Conclusion.** Liquidity frictions do not just reduce the volume of capital reallocation; they change its composition from efficient full transfers toward inefficient partial transfers.
- **Intuition.** Inflation taxes liquidity in the long run, so buyers arrive at the secondary market

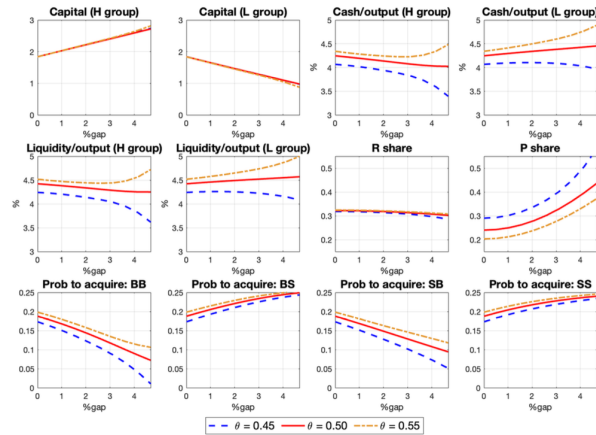


FIG. 11.—Effects of differences in persistent productivity component.

with less spending power. In the short run, easier credit can both increase reallocation and show up as higher inflation by lowering money demand.

- **Empirical lesson.** To understand used-capital markets, we need both micro evidence on firms' liquid balance sheets and a structural framework that distinguishes money, credit, search, and bargaining.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** Higher long-run inflation reduces total capital reallocation, reduces full sales, and raises the share of partial sales. At business-cycle frequencies, adding credit shocks reverses the sign in exactly the way seen in the data.
- **Mechanism.** The key friction is endogenous liquidity. Buyers need cash and pledgeable collateral to acquire capital; when those are scarce, they cannot buy the whole bundle even if it is efficient for them to do so.
- **Quantitative success.** The calibrated monetary model matches standard RBC facts *and* the cyclicalities of the R share, P share, inflation, used-capital prices, and productivity dispersion.
- **Policy insight.** Inflation is costly not only because it distorts money demand, but also because it obstructs capital reallocation. In addition, bargaining can make a capital subsidy desirable, and the Friedman rule is not automatically optimal in distorted environments.

7.2 Economic intuition

- In a frictionless asset market, capital moves to the highest-productivity firm in one shot. In this paper, that same logic holds as a social benchmark.
- What blocks the benchmark is not disagreement about where capital should go; it is the lack of purchasing power at the moment of trade.
- That is why money and collateral matter for TFP. They determine whether a productive firm can finance the whole acquisition or only a slice of it.
- Credit shocks are especially powerful at business-cycle frequencies because they simultaneously relax acquisition constraints and alter money demand.

7.3 Takeaway

Macro, money, and asset-market frictions. Its biggest conceptual contribution is to turn the distinction between *full* and *partial* sales into a measurable object that reveals the severity of liquidity distortions. The empirical evidence, analytic structure, and calibration all line up around the same message: liquidity is a central determinant of how efficiently capital gets reallocated across firms.